Smart Modular Switch System

Make every switchboard in India intelligent — without rewiring, without overpaying, without trapping users in one company's app.

₹1 Cr	48%	9 mo
pre-seed ask	cheaper than Wipro Smart for 8 channels	to MVP launch, electrician-channel only

Pre-seed pitch · May 2026 · Confidential

The Indian smart switch market is broken in two ways.

Penetration is under 5% in residential — not because demand is missing, but because the product is.

Expensive incumbents Wipro, Havells, Syska, Legrand	**Cheap imports** Tuya-based clones, Amazon brands
₹1,200–1,500 per gang. A typical 8-switch home costs ₹10,000+ just for the switches. Add an electrician's wiring time and you've spent ₹15,000 to make 8 lights smart. **One Wi-Fi radio in every switch.** Every switch pairs separately, every one is a failure point, every one needs the home Wi-Fi to work.	**Cloud-dependent.** Stop working when the company changes its API or shuts down (many already have). **Sluggish, no local control.** 2–4 second response from touch to relay because every command round-trips through Shanghai. Customers churn fast. **No mesh, no inter-switch awareness.** Every switchboard is an island.

The pattern: the budget end is unusable; the usable end is unaffordable. 30 million Indian households want this and can't get it.

Stop putting a Wi-Fi radio in every switch.

One Master with all the radios. Up to 5 dumb-but-cheap Extensions hanging off a low-voltage bus.

Master module (₹775 BOM)

ESP32-C6 · Wi-Fi 6 · BLE 5 · 802.15.4 · 2 relays · energy monitoring · RTC

Extension module (₹308 BOM)

CH32V003 · no radios · 2 relays · powered from bus · plug-and-play

Snap-Link bus

RS-485 · 250kbps · auto-discovery · 5 extensions per master

Mesh

Multiple masters across a home form a single network

ESP-MESH-LITE · up to 8 masters · works without internet

What changes
Radios per home (8 switches): Wipro: 8 radios · Us: 1 radio
Customer price for 8 channels: Wipro: ~₹9,600 · Us: ~₹5,000
Install time per switchboard: Wipro: 45+ min · Us: under 20 min
Failure points per home: Wipro: 8 Wi-Fi pairings · Us: 1 per board

One radio across 12 channels. The architecture itself is the cost advantage.

What we ship in v1.0.

Same hardware ships for v1, v2, and v3. New features unlock over OTA. Customers never replace their switches to upgrade.

v1.0 — Local Mesh	v2.0 — Remote Control	v3.0 — Open Ecosystem
Months 0–9 · MVP	Months 9–15	Months 15–24
App-controlled locally over Wi-Fi or Bluetooth **Mesh across multiple masters** in same home — any switch from any room **Schedules, scenes, energy monitoring** — all working offline **<30ms touch response** regardless of network state **No internet required** for any function	**Remote control** from anywhere via the cloud **Push notifications** — door switched, energy threshold, master offline **Multi-user access** with permission levels **Energy history** + cloud-side automations (sunrise/sunset, geofencing) **Pushed to v1 hardware over OTA** — no replacement needed	**Matter over Wi-Fi** — Apple Home, Google Home, Alexa, SmartThings **Thread border router** — each master anchors a Thread network **Local LAN API** for Home Assistant + power users **Pushed to v1 + v2 hardware** **Same hardware. New capabilities.**

The Tesla story applied to switches. Buy v1 today. Get v2 features in a year, v3 in two. Same warranty. Same hardware. New capabilities, free, over the air.

Mesh is the differentiator. Nobody in the budget segment has it.

Cheap smart switches treat every switchboard as an island. We mesh them.

Without mesh (every cheap switch today)	With our mesh
User has 4 switchboards across a 3-BHK. To control switch in living room from bedroom, they need to: • Be in Wi-Fi range of that specific switch • Or call out to Alexa/Google (cloud round-trip, 2-4s) • Or get up and walk over **4 switchboards = 4 disconnected islands.**	All 4 masters in the home form a single network. The user's app talks to whichever master is nearest. That master forwards the command to whichever master owns the target switch. • **Works locally** — no internet needed • **Works through Wi-Fi outages** — masters mesh peer-to-peer • **One pairing per home**, not per switchboard • **One Wi-Fi password change** propagates to all masters

Technical defensibility: ESP-MESH-LITE over Wi-Fi, secured by 128-bit per-home mesh key rotated on device-leave. Each device manufacturer-signed by our private key (in HSM) so clones cannot join the mesh. Provisional patents filed on the master-swap and re-adoption flows.

₹1,40,000 crore addressable in 5 years.

30 million mid-income Indian households + new construction. 1% share is ₹1,400 Cr revenue.

TAM	SAM	SOM at year 3
₹1,40,000 Cr 30M households · 8 switches/home · ₹600 avg India's mid-income urban + new construction smart switch retrofit opportunity over the next 5 years.	**₹14,000 Cr** Top-20 metros · electrician-installed · mid-income The segment we can actually reach with our channel strategy (electricians, then D2C). 10% of TAM.	**₹140 Cr** 100,000 active homes · 8 channels each Our 3-year north star. 1% of SAM, ~0.1% of TAM. Achievable with seed proceeds + Series A.

Why now

1. Smart home awareness has crossed the chasm. Alexa and Google Home in 12M+ Indian households. Consumers know smart-home is possible — they just can't afford it at current price points.

2. Wi-Fi 6 + Matter just made the silicon affordable. ESP32-C6 (~₹225 at MVP volume, ₹150 at scale) puts Matter-capable radios in budget products for the first time.

3. Indian housing supply is up. Tier-2 city construction is the largest growth driver. New homes = greenfield switch installs, not retrofits. We capture both.

4. The incumbents are asleep. Wipro/Havells haven't refreshed product in 3+ years. Tuya brands are losing ground on cloud-dependence concerns. The window is open.

Hardware margins like consumer electronics, retention like SaaS.

60%+ margin at street price. OTA upgrade story creates Tesla-like brand loyalty in a category that has none.

SKU	Street price	BOM	Margin	Comparable
Master (2-gang) Wi-Fi, BLE, energy meter, RTC, mesh	₹1,999	₹775	**~58%**	Wipro 2-gang Smart: ₹2,400, single-radio
Extension (2-gang) Bus-powered, no radio, daisy-chain	₹999	₹308	**~65%**	No direct competitor at this price point
Typical 8-channel home (1+3)	**₹5,000**	**₹1,700**	**~66%**	**Wipro equivalent: ~₹9,600 (48% premium)**

Phase 1 — Electricians Months 0–9	Phase 2 — DIY / Amazon Months 6–18	Phase 3 — Builders Months 12–24
80% of Indian switch purchases are decided by the electrician. We train and equip them with starter kits, commissions, and a WhatsApp support group. **Target M9:** 50–100 trained electricians in anchor city, 200 paying installs.	D2C via Amazon and Flipkart once we have install reviews, BIS certification, and the cloud features from v2. **Target M18:** 10,000 D2C orders, 3-city presence.	Tier-2 developers first (tier-1 wants established brands). Specifier-grade datasheets, BIS copies, project case studies, in-person sales. **Target M24:** 3 projects of 500–2,000 units each.

₹1 Cr pre-seed buys validation, not launch.

Working hardware + 200+ paying installs + BIS certification underway → ready to raise ₹3–4 Cr seed at Month 9 against demonstrated traction.

Spend area	₹L	What it produces
Founding team salary (3 × 7 months)	35	Working prototype, firmware, app v1
Hardware development (3 board revs, mesh kits)	12	Validated Master + Extension + mesh
Pilot inventory (50 units at CM minimum)	18	Real installs in 20+ homes
BIS + WPC + pre-compliance EMC	6	Certification underway, no surprises pre-launch
Tools & lab (iso transformer, scope, jigs)	4	Bench capable of mains-side bring-up
Electrician training program (anchor city)	8	50 trained installers, channel seeded
Legal, accounting, IP filings	3	Provisional patents on bus + mesh adoption
App development (Flutter, contract)	6	v1 app with mesh-aware UX
Contingency (8%)	8	Hardware buffer for inevitable slip
Total pre-seed	**100**	

The four things we prove with ₹1 Cr (so seed investors say yes at Month 9):

1. It works. Master + 5 extensions + 2-master mesh all functional. Thermal soak passed. EMC pre-scan clean.
2. Electricians install it. 50+ trained installers, 200+ paid installs, install time under 20 min.
3. Customers like it. 200+ installs running 30+ days, NPS >50, churn / RMA <5%.
4. The mesh story holds up. 20+ multi-master homes in active use, logged data showing peer-to-peer operation through Wi-Fi outages.

What can go wrong — and what we do about it.

Hardware startups fail in predictable ways. We've planned for each.

Risk	How we mitigate
BIS certification delays launch. Indian safety cert routinely takes 6+ months and can block ship dates.	File paperwork on Day 1, not when 'hardware is ready'. Lab booking timeline starts before our hardware does. WPC + EMC parallel-tracked.
Bad OTA bricks the fleet. Every master is in someone's wall. A bad update sent to 10,000 units is catastrophic.	Staged rollout (max 1% in 24h, auto-pause on crash spike). Dual-partition with 3-failure auto-rollback. BLE-only recovery mode via recovery pin.
Master failure dark-rooms the whole switchboard. If the master dies, every extension on its bus stops responding.	Local fallback on extensions — touch keeps working even when master is offline. Master swap procedure: new master adopts orphan extensions automatically.
Electrician channel slower to ramp than projected. Channel development is unpredictable, especially in a single city.	Direct D2C via Amazon as parallel channel from Month 9 onwards. Hardware works regardless of channel; can pivot pilot to D2C-only if needed.
Counterfeit clones at scale. Successful hardware in India invites Chinese clones within 18–24 months.	Mesh authentication makes cloning useless without our manufacturer root key. Bus protocol IP filed as provisional patent. Electrician brand trust as moat.
ESP32-C6 supply shortage. Single-source on critical silicon is always a risk.	Pin-compatible fallback design with ESP32-C3 (loses Thread for v3 only, keeps Wi-Fi+BLE+mesh). Acceptable v1 emergency fallback.
Pre-seed runs out before pilot data lands. Hardware projects routinely slip 30%+.	8% contingency built in. Pre-defined scope cuts (master-only, no mesh demo) if we hit Month 5 without working hardware. Bridge raise if any traction at all.

₹1 Cr pre-seed. Working product + paying customers in 9 months.

What we're raising

₹1 Crore

pre-seed, equity, valuation TBD on conversation

Use of funds: 35% team, 12% hardware development, 18% pilot inventory, 8% electrician channel, 6% certification, 6% app, 4% lab, 3% legal/IP, 8% contingency.

Runway: 7–9 months to working prototype + 200+ paying installs + BIS underway.

Next round: ₹3–4 Cr seed at Month 9, on demonstrated traction.

What you're investing in

A product 30 million Indian households want, priced 48% below the cheapest reliable alternative, architected so customers never replace it.

A wedge — mesh — that incumbents cannot easily copy without redesigning their product line, and that import clones cannot copy without the manufacturer-root-key security that makes mesh trustworthy.

A channel — the Indian electrician — that nobody is investing in, even though they decide ~80% of switch purchases in this country.

A team that has thought through the hard parts (mesh credential rotation, OTA rollback, master swap re-adoption, certification timing) before writing a line of firmware.

We are raising to prove the product works and that users want it — so we can raise the real round on what we built, not what we said.

Contact: [founder name] · [email] · [phone] · Companion materials: Handover v1.1, Engineering v1.0, BOM/order list.